

Poisson Distribution

A discrete r.v. X is said to follow Poisson distribution if the pmf of the r.v. X is given by

$prob(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$, where $k=0,1,2,3,\dots$ and λ is the parameter of the distribution. It is denoted as $X \sim Poi(\lambda)$

Following are some instances when Poisson distribution can be successfully used.

1. Number of death cases from a disease like: heart attack, cerebral attack, etc.
2. Radio active decay
3. Number arrival in telephone booth, in bank, etc.\
4. Number of accidents in a specific unit time
5. Number of suicide reported in particular city, etc.

The Poisson distribution is an appropriate model if the following assumptions are true:^[5]

1. The number of trials k is the number of times an event occurs in an interval and k can take values 0, 1, 2,
2. Trials are independent.
3. The average rate at which events occur is independent of any occurrences and it is constant.

Properties of Poisson distribution

1. mean and variance of Poisson distribution is same and it is equal to the parameter λ
2. If X_1 and X_2 are two independent Poisson r.vs. with parameters λ_1 and λ_2 the sum of the two r.vs. i.e., $X_1 + X_2$ will follow Poisson distribution with parameters $\lambda_1 + \lambda_2$. This can be extended for n independent Poisson random variables. This is known as additive property.

Mean and variance of Poisson distribution:

Mean of Poisson distribution =

$$\mu = E(X) = \sum_{k=0}^{\infty} k \times prob(X = k) = \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!} = \lambda e^{-\lambda} + 2 \frac{\lambda^2 e^{-\lambda}}{2!} + \dots = \lambda e^{-\lambda} (1 + \lambda + \frac{\lambda^2}{2!} + \dots) = e^{-\lambda} \lambda e^{\lambda} = \lambda$$

Similarly the variance is $\sigma^2 = \lambda$ The moment generating function is mgf =

$$M_X(t) = \sum_{k=0}^{\infty} e^{tk} prob(X=k) = \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} + \frac{e^t \lambda e^{-\lambda}}{1!} + e^{2t} \frac{\lambda^2 e^{-\lambda}}{2!} + \dots = e^{-\lambda} (1 + \lambda e^t + \frac{\lambda^2 e^{2t}}{2!} + \dots) = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}$$

Hence mean is $E(X) = M'_x(t)$ at $t=0$ is λ and variance is $\sigma^2 = M''_x(t) - \{M'_x(t)\}^2$ at $t=0 = \lambda$

Poisson distribution as an Approximation to the Binomial Distribution:

- Conditions: (i) n , the number of trials is indefinitely large.
 (ii) p , the constant probability of success for each trial is indefinitely small, i.e. $p \rightarrow 0$, $n \rightarrow \infty$
 (iii) $np = \lambda$ (say) is finite. $p = \frac{\lambda}{n}$, $q = 1 - \frac{\lambda}{n}$

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x} = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

Let $np = \lambda$ then

$$P(X=x) = \frac{n(n-1)(n-2) \cdots (n-x+1)(n-x)!}{x! (n-x)!} p^x (1-p)^{n-x}$$

$$= \frac{n(n-1)(n-2) \cdots (n-x+1)}{x!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{n(n-1)(n-2) \cdots (n-x+1)}{x! n^x} \lambda^x \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x}$$

$$= \frac{n}{n^x} \left(1 - \frac{\lambda}{n}\right) \left(1 - \frac{\lambda}{n}\right) \cdots \left(1 - \frac{\lambda}{n}\right) \cdot \frac{\lambda^x}{x!} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x}$$

$$= 1 \cdot \frac{e^{-\lambda} \lambda^x}{x!} \left[\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \rightarrow 1 \text{ as } \frac{\lambda}{n} \rightarrow 0 \right]$$

$$= \frac{e^{-\lambda} \lambda^x}{x!}$$

Example

1. Suppose that X has a Poisson distribution. If

$\text{Prob}(X = 2) = \frac{2}{3} \text{Prob}(X = 1)$, evaluate i). $\text{Prob}(X = 0)$ and ii). $\text{Prob}(X = 3)$